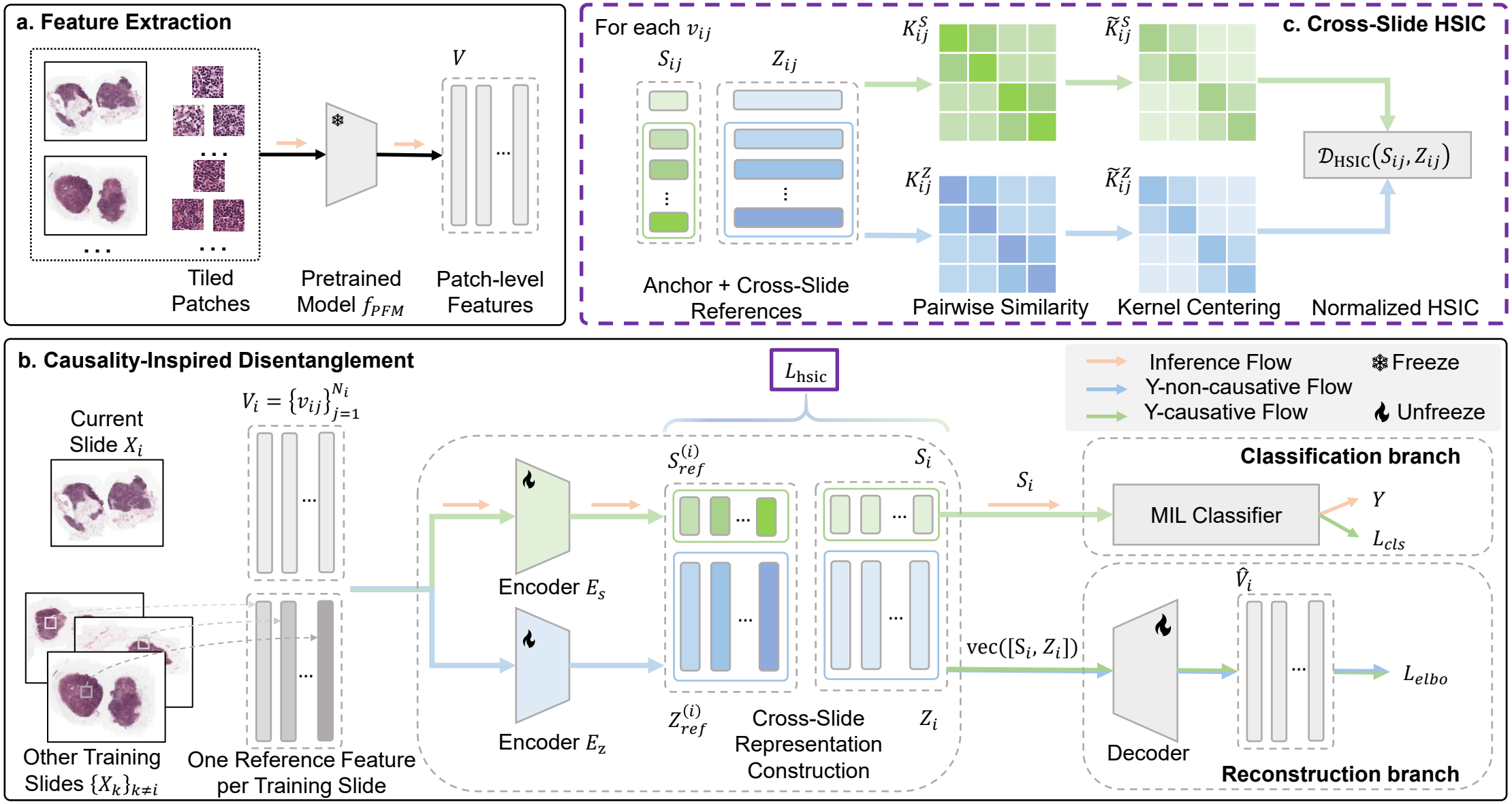

a. Feature Extraction
Tiled Patches
Pretrained Model f_{PFM}
V
Patch-level Features
c. Cross-Slide HSIC
For each v_{ij}
S_{ij}
Z_{ij}
K_{ij}^S
$\tilde{K}_{ij}^S$
K_{ij}^Z
$\tilde{K}_{ij}^Z$
$\mathcal{D}_{\text{HSIC}}(S_{ij}, Z_{ij})$
Anchor + Cross-Slide References
Pairwise Similarity
Kernel Centering
Normalized HSIC
b. Causality-Inspired Disentanglement
L_{hsic}
Inference Flow
Y-non-causative Flow
Y-causative Flow
Freeze
Unfreeze
Current Slide X_i
$V_i = \{v_{ij}\}_{j=1}^{N_i}$
Other Training Slides $\{X_k\}_{k\neq i}$
One Reference Feature per Training Slide
Encoder E_s
Encoder E_z
$S_{ref}^{(i)}$
$Z_{ref}^{(i)}$
Cross-Slide Representation Construction
S_i
Z_i
Classification branch
MIL Classifier
Y
L_{cls}
$\text{vec}([S_i, Z_i])$
Decoder
$\hat{V}_i$
L_{elbo}
Reconstruction branch